

Samaksh Rastogi

Software Engineer | Full Stack & Backend Developer

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OVERVIEW

Software Engineering Intern at Nokia with 11 months of experience building enterprise-scale web applications, automation platforms, analytics systems, and AI-powered solutions. Experienced in Python, Django, Node.js, React, AWS, and distributed architectures, with hands-on contributions to media processing platforms, network operations systems, and workflow automation tools. Passionate about backend engineering, scalable software design, and full-stack product development.

SKILLS

Backend: Django, Django REST Framework (DRF), Node.js, Express.js, RESTful APIs, Authentication & Authorization, OAuth 2.0, Role-Based Access Control (RBAC)

Languages: Python, JavaScript, TypeScript, SQL

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3, Responsive Web Design

Database & Cache: PostgreSQL, MongoDB, Redis, SQLite, Database Design, Query Optimization

Cloud & DevOps: AWS S3, AWS CloudFront, Azure AD SSO, Vercel, Render, Linux, CI/CD Fundamentals, Deployment & Hosting

AI and Automation: LangChain, LangGraph, Ollama, AI Agents, RAG (Retrieval-Augmented Generation), Prompt Engineering, Workflow Automation

PROFESSIONAL EXPERIENCE

Nokia Solutions and Networks India Pvt. Ltd., *Student Intern*

08/2025 – 06/2026

Gurugram, India

- Contributed to three enterprise-scale Nokia platforms supporting media management, network operations, workflow automation, analytics, and issue management systems.
- Nokia Inventory Video Repository (NIVR)**
 - Developed Nokia's enterprise video repository with Azure AD SSO, RBAC, secure approval workflows, and 50+ application routes supporting content management, analytics, and governance.
 - Built AI-powered media processing pipelines using FFmpeg, OpenCV, Faster-Whisper, spaCy, and sentence-transformers for automated transcription, thumbnail generation, metadata extraction, and intelligent search.

Technologies: Python, Django, Azure AD SSO, AWS S3, FFmpeg, OpenCV, Faster-Whisper, spaCy
- Simplified Technical Support Interface (STSI)**
 - Developed an enterprise support portal centralizing health checks, RCA workflows, log collection, and 10+ operational utilities to improve network operations productivity.
 - Built scheduling, automation, and analytics services using Celery, APScheduler, Pandas, Plotly, and Matplotlib for recurring operations and reporting dashboards.

Technologies: Django, DRF, Celery, APScheduler, Pandas, Plotly, Matplotlib
- Hot Issues Dashboard (TSHID)**
 - Developed REST APIs, approval workflows, and notification systems for enterprise issue governance, enabling streamlined customer and product incident management.
 - Implemented dashboards, reporting modules, exports, and email escalation workflows to improve issue visibility, resolution tracking, and stakeholder communication.

Technologies: Django, DRF, REST APIs, Celery, APScheduler, Email Notifications

PROJECTS

SK Central , <i>Centralized Identity (SSO), Real-Time Operational Telemetry, and AI Governance Platform</i> - Architected a zero-trust SSO platform using HttpOnly cookies, 15-minute JWTs, MongoDB, and Redis, achieving <50 ms authentication latency, <1 ms session lookups, and reducing database load by 80%. - Built a distributed event-driven platform with Socket.IO , Redis Pub/Sub, circuit-breaker fault isolation, and a Gemini-powered RAG assistant, maintaining 99.9% uptime while optimizing the React SPA bundle to <150 KB. Tech Stack: React 19, TypeScript, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Redis, Socket.IO , Google Gemini API, Docker. Live	04/2026 – Present
SK Quiz Coach , <i>An AI-driven adaptive exam preparation and mentoring platform.</i> - Developed an AI-powered adaptive learning platform featuring dynamic difficulty adjustment, spaced repetition, and Gemini 2.5 Flash pipelines for structured exam content generation with 30+ validated schema fields. - Engineered a scalable TypeScript monorepo with Redis, BullMQ, and background job orchestration for reliable notifications and high-performance backend workflows. Tech Stack: React 19, Node.js, Express, TypeScript, MongoDB, Redis, BullMQ, Google Gemini API, TanStack Router/Query, Zustand, Socket.io , Tailwind CSS, Recharts, Framer Motion, Docker. Live	02/2026 – 05/2026
SK MailPilot , <i>AI-powered Gmail workspace for inbox triage, smart replies, and workflow automation</i> - Built an AI-powered Gmail workspace using LangGraph, LangChain, Gemini, and Google OAuth to automate email classification, smart replies, and workflow orchestration across multiple priority levels. - Designed a scalable backend with Redis, BullMQ, Gmail APIs, and a RAG-powered assistant to support batch synchronization, enterprise RBAC, real-time audit trails, and conversational inbox management. Tech Stack: React 19, TypeScript, Node.js, Express 5, LangGraph, LangChain, Gemini API, Ollama, MongoDB, Redis, BullMQ, Gmail API, Google OAuth. Live	04/2026 – 04/2026
SK Connect , <i>Connect effortlessly with real-time text, P2P video calls, expiring status stories, and smart AI insights.</i> - Built real-time messaging & WebRTC video calling using Socket.IO and Redis adapters with <100ms latency across 3 receipt states. - Developed Discord-style communities with 5+ channel types, 4-digit pairing codes, and MongoDB TTL-backed 24-hour expiring stories with self-destruct timers (5s–24h). - Integrated Gemini AI for contextual chat summaries and live polls syncing real-time updates across 100+ concurrent users in a Dockerized monorepo. Tech Stack: React 19, TypeScript, Node.js, Express, Socket.IO , WebRTC, MongoDB, Redis, Gemini AI, Docker. Live	05/2026 – Present

EDUCATION

Bachelor of Technology in Information Technology , <i>Chandigarh Engineering College, CGC Landran</i> CGPA: 8.04/10	08/2022 – 06/2026 Punjab INDIA
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CERTIFICATES

- Introduction to Artificial Intelligence (Nokia)
- Cloud Computing (Nokia)
- NFV & Telco Cloud (Nokia)
- Machine Learning (NPTEL, IIT Madras)